

INFRARED MOTION-SENSING Relay Switch

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This circuit is designed for use with all kinds of medium-power automobile/domestic 12V DC loads. It is a simple solid-state relay (SSR) switch, controlled by a standard passive infrared (PIR) motion sensor module. A PIR sensor is an electronic device that can measure IR light radiating from objects in its field-of-view.

Apparent motion is detected when an IR source with one tem-

perature (such as a human being) passes in front of an IR source with another temperature (such as a wall). The PIR sensor module, centred on a PIR sensor, has elements made of crystalline material that generates an electric charge when exposed to IR radiation.

Changes in the amount of IR striking the element change the voltages generated, which are measured by an onboard circuitry.

The module contains a special filter called Fresnel lens, which focuses IR signals onto the sensor element. As ambient IR signals change rapidly, onboard circuitry triggers the output to indicate motion. Fig. 1 shows the PIR motion sensor module. It is usually hidden behind a translucent dome shaped cover as shown in the inset in Fig. 1.

Circuit and working

Circuit diagram of the PIR motion-sensing SSR switch is shown in Fig. 2. It is built around a PIR motion-sensor module (connected across CON2), MOSFETs BS170 (T1) and IRF9540 (T2), rectifier diode 1N4007 (D1) and a few other components.

The PIR sensor module has a 3-pin connection for Vcc, output and ground, and provides a single output that goes high when motion is detected. It also has a 2-pin jumper selection for single or continuous trigger output mode. The two positions are labelled H and L (Fig. 1).

When the jumper is at H position, output goes high whenever the sensor is triggered and retriggered.

In position L, output goes high once and then low like a monostable timer every time the sensor is triggered. Retriggering pulse does not affect the output duration.

The circuit presented here is in repeatable trigger mode and turns T2

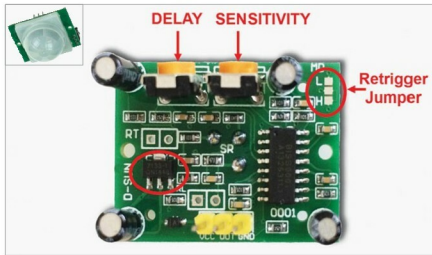


Fig. 1: PIR motion sensor module and its dome shaped cover (inset)

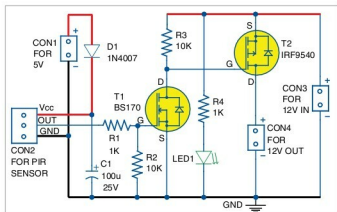


Fig. 2: PIR motion Sensing SSR Switch

PARTS LIST

Semiconductors:

- D1 - 1N4007 rectifier diode
- T1 - BS170 n-channel MOSFET
- T2 - IRF9540 p-channel MOSFET
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R4 - 1-kilo-ohm
- R2, R3 - 10-kilo-ohm

Capacitor:

- C1 - 100µF, 25V electrolytic

Miscellaneous:

- CON1, CON3, CON4 - 2-pin connector terminal
- CON2 - 3-pin connector
- PIR motion sensor